

# Shashank S

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Machine Learning Engineer with 6 years shipping real-time recommendation and ML platform systems at 100M+ DAU scale; High Performance Computing and Analog Compute-in-Memory at Georgia Tech.

## EXPERIENCE

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### Senior Machine Learning Engineer, Walmart | *Dec 2024 – Jul 2025*

- Transitioned to full-time MS study.
- Cut compute spend growth on the adtech ML pipeline by profiling PySpark bottlenecks, rewriting shuffle-heavy transformations, and scaling batch processing to keep up with advertising volume growth.
- Accelerated business and ML pipeline analytics for non-technical teams by building a Text2SQL agentic system with plain-English querying, visualizations, and active chat for self-serve insights.
- Improved production ML reliability and release velocity by building an observability framework and standardizing model build, test, and deployment workflows with GitHub Actions.

### Machine Learning Engineer III, Glance-InMobi | *May 2021 – Nov 2024*

- Delivered personalized content to **100M DAU** at **sub-50ms p99 latency** by building low-latency ML serving and streaming pipelines with prediction services, Vertex AI feature store, Kafka, ELK logging, and Airflow retraining.
- Increased engagement by **5%** and conversion by **12%** with recommendation models including multi-armed bandits, Elo ratings, GBMs, and graph-based collaborative filtering.
- Boosted time spent by **11.68%** overall and **18.38%** for cold-start users by co-authoring lookalike modeling research using SVD, K-Means, and bandit-based exploration.
- Owned ML automation and experimentation workflows for streaming pipelines, model controllers, rollback triggers, and event-driven personalization, improving release reliability for recommender launches.
- Built automated quiz generation over live video streams in real time, for YouTube Live Streams.

### Data Scientist, Bright Money | *Jun 2019 – Apr 2021*

- Founded the data science function at an early-stage fintech and built ML systems for debt management, cash-flow forecasting, affordability scoring, and overdraft prediction.
- Launched *Debt Manager* on Airflow and Django, driving adoption to **10K+ users** and **\$100K+ daily transaction volume** through automated financial analysis and payment scheduling.
- Developed bill identification, income forecasting, expense forecasting, and account-balance prediction models for product retention and risk workflows.

## EDUCATION

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### Master of Science, Computer Science, Georgia Institute of Technology | *May 2026*

- Specialization: High Performance Computing; GPA: 3.87/4.0.
- Research: Analog compute-in-memory error mitigation with Prof. Hyesoon Kim
- Coursework: GPU Hardware & Software, High-Performance Parallel Computing, HPC Architecture, Scientific ML, Deep Learning, Designing for Curiosity.

### Bachelor of Engineering, Computer Science, Sir M. Visvesvaraya Institute of Technology | *May 2018*

- First Class with Distinction.
- Thesis: Image Regeneration with Generative Models; used CapsNet as a GAN discriminator for semantic inpainting on MNIST and face images.

## SELECTED SYSTEMS & RESEARCH

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**Embedded Drift Sentinel** (ASPLOS 2027 submission): Improved day-360 ImageNet **top-1 accuracy by +14 pp** on average over GDC across ResNet-18/34/50 and GoogLeNet on a 360-day PCM drift horizon (ResNet-50: 68% vs. 53% GDC, vs. 17% uncorrected). Designed a companion-crossbar architecture for online drift correction in analog compute-in-memory accelerators; eliminates main-array probe downtime at 0.3–1.7% cell overhead.

**Physics-Informed RL for Plasma Control:** Trained PPO policies in TORAX (auto-differentiable 1D core transport simulator) on a 50-parameter partially-observed state for ITER tokamak fusion control; achieved **1.7 ms average inference** (3.5 ms max), well under the 50 ms real-time threshold.

**Meibomian Gland Segmentation:** Published CNN-based meibography segmentation work in *Biomedical Signal Processing and Control*; developed custom augmentation and gland-health metrics validated against clinical expert annotations across tabletop and prototype handheld imagers.

**Automated Debt Management Patents:** R&D contributor; [US20220261886A1](#), [US20230075411A1](#).

## HONORS & TEACHING

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**ACM ICPC 2015:** Qualified to regional level; honorable mention.

**Section Leader, Code in Place, Stanford University:** Taught Python to 30+ students per cohort in [2021](#), [2023](#), and [2024](#).

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, PySpark, SQL, Go, Shell

**ML Systems:** PyTorch, scikit-learn, XGBoost/LightGBM, Deep Learning, Recommendation Systems, LLMs, LangChain, Agentic Workflows, MLOps, A/B Testing

**HPC & AI Infrastructure:** CUDA, MPI, OpenMP, GPU Profiling - Nsight Systems and nvprof, Distributed Systems, Scientific ML, Spark, Kafka, Docker, Kubernetes, Airflow, Argo, Vertex AI, Prometheus, Grafana

**Cloud & Data:** GCP, AWS, Azure, PostgreSQL, Django, Linux, Git